

ECON 144 — FULL PRACTICE FINAL EXAM SECTION 1: Time Series Fundamentals 1. Define weak stationarity. 2. Interpret sample ACF and PACF plots. SECTION 2: Trend/Seasonality 3. Fit a linear trend model and interpret coefficients. 4. Explain additive vs multiplicative seasonality. SECTION 3: ARIMA Modeling 5. Identify AR and MA orders from ACF/PACF. 6. Explain differencing and unit roots. SECTION 4: ARCH/GARCH 7. Compute persistence. 8. Interpret GARCH(1,1) forecasts. SECTION 5: Forecast Evaluation 9. Run a DM test and interpret. 10. MZ regression: test optimality. SECTION 6: Cointegration & ECM 11. Apply Engle–Granger. 12. Interpret ECM coefficients. SECTION 7: Kalman Filter 13. Explain Q, R, and Kalman Gain. 14. Time-varying beta filtering interpretation. SECTION 8: MCMC & Markov Chains 15. Gibbs sampling steps. 16. Stationary distribution of a 2x2 Markov chain.